

The L1 figure uses Experiment 1’s whole-tensor canonical layouts, stratified over all score components before profiling. The H100 L1 comparison uses the issue quotient Q_{32B} , matching the counter’s 32-byte sector unit.¹ It gives perfect rank agreement for all six kernels with layout variation; correlations with downstream L2 activity are less uniform.

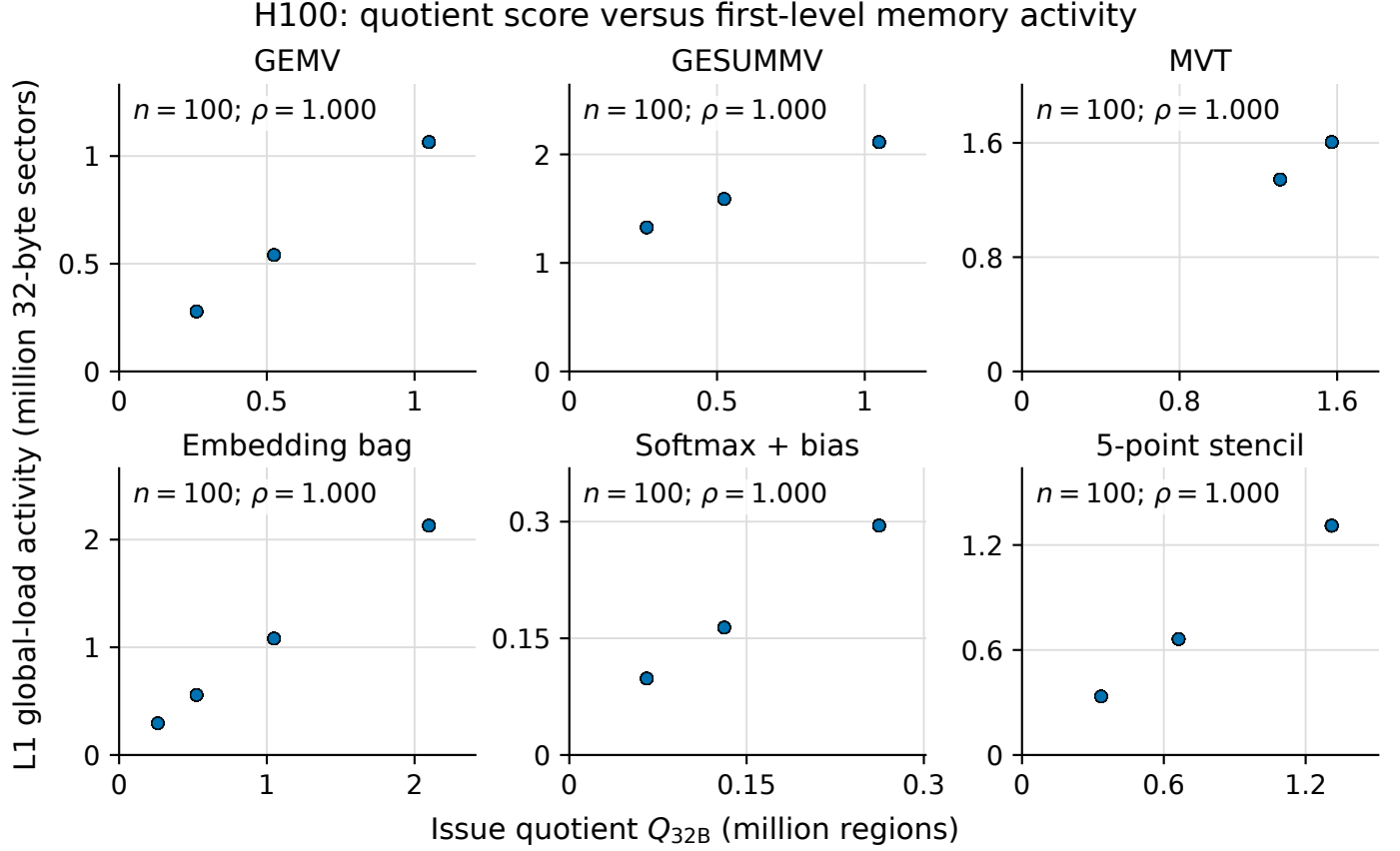


Figure 1: Issue quotient versus first-level memory activity for all six pilot kernels on NVIDIA H100. The horizontal axis is the target operand’s dynamic issue quotient Q_{32B} (component `issue.g32.stream.load.32B`); the vertical axis is whole-kernel global-load L1TEX sectors, measured by `l1tex__t_sectors_pipe_lsu_mem_global_op_ld.sum`. Both axes use millions and start at zero; panel scales differ. Each point is one distinct whole-tensor canonical mapping from the all-scope-stratified Experiment 1 panel. Repeated coordinates overlap; n counts mappings, not visible markers. Points are medians of three independent profiler-launch medians, each over 20 steady-state dispatches under the warm-cache protocol. All three launch medians coincide for every mapping, so error bars have zero height. Spearman’s ρ uses average ranks for ties. The quotient scores only the varied operand, whereas the counter includes all kernel loads; offsets therefore need not vanish.

¹NVIDIA Nsight Compute, Profiling Guide: metrics and memory workload analysis.

The L2 comparison uses Experiment 2’s canonical inner-tile layouts, stratified across all score components, with row-major outer tiles. We correlate J_{area} under the device-specific `l1_to_l2` tau profile with read requests sent from L1 to L2.

Kernel	H100	MI300A
GEMV	0.299	0.977
GESUMMV	0.442	0.727
MVT	0.771	0.935
Embedding bag	0.966	0.988
Softmax + bias	0.959	1.000
5-point stencil	0.931	0.999
Median	0.851	0.982

Table 1: Spearman correlations between J_{area} and L1-to-L2 read requests on H100 and MI300A, using the `l1_to_l2` tau profile and Experiment 2’s all-scope-stratified canonical inner-tile layout panels. Each per-kernel correlation uses 100 distinct mappings and average ranks for ties. H100 counts TEX-source L2 read requests with `lts__t_requests_srcunit_tex_op_read.sum`; MI300A counts TCP-to-TCC read requests with `TCP_TCC_READ_REQ.sum`. Each counter observation is the median of three independent profiler-launch medians, each over 20 steady-state dispatches with warm caches. The final row is the median of the six per-kernel correlations, not a pooled correlation or a median across launches. Bias+ReLU is excluded. The device-specific tau weights were fitted using pilot counters; these are pilot-panel correlations.